FN1 Worked Examples: Is It a Function?

Recording companion for the FN1 CheckIt support video — Math 172, Santa Ana College

How to use this document

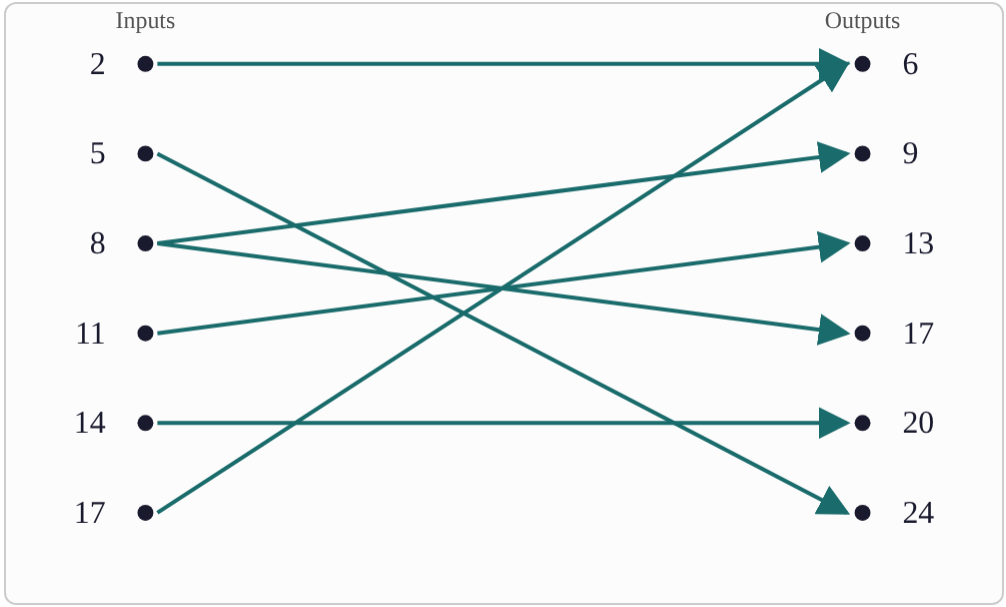
- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Can one input ever correspond to more than one output? If yes, it is not a function.

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Example 1: Mapping Diagram



Plotted data: 2 → 6, 5 → 24, 8 → 9, 8 → 17, 11 → 13, 14 → 20, 17 → 6.

Question

Does this mapping define a function?

Answer and reasoning

The input **8** appears twice in the list — once going to 9, once going to 17. One input can't correspond to two different outputs, so **this mapping does not define a function.** (Every other input appears only once, so that one repeat is the whole story.)

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Example 2: Table

Table 1. Table of x-values and y-values, listed in two rows

x	4	7	10	13	16	19
y	11	2	15	8	3	9

Question

Does this table define a function?

Answer and reasoning

Scan the x-row: 4, 7, 10, 13, 16, 19 — every value appears exactly once. No input is repeated, so **this table defines a function.** (It doesn't matter that some y-values might repeat — only repeats in x break the definition.)

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Example 3: Ordered Pairs

(3, 9), *(3, 14)*, *(6, 2)*, *(10, 5)*, *(15, 8)*, *(18, 1)*

Question

Do these ordered pairs define a function?

Answer and reasoning

(3, 9) and *(3, 14)* share the same x-value, 3, but disagree on y. That's the same conflict as a repeated input in a mapping diagram — just written as coordinates. So **these ordered pairs do not define a function.**

Example 4: Equation 1

$$3y^5 + 2x^4 = 0$$

Question

Does this equation define a function of x?

Answer and reasoning

Solving for y here means taking a **fifth root** — an odd root. Odd roots are invertible, with no sign ambiguity: each real number has exactly one real fifth root. So **this equation defines a function of x.**

Example 5: Equation 2

$$2y^4 + 3x^5 = 0$$

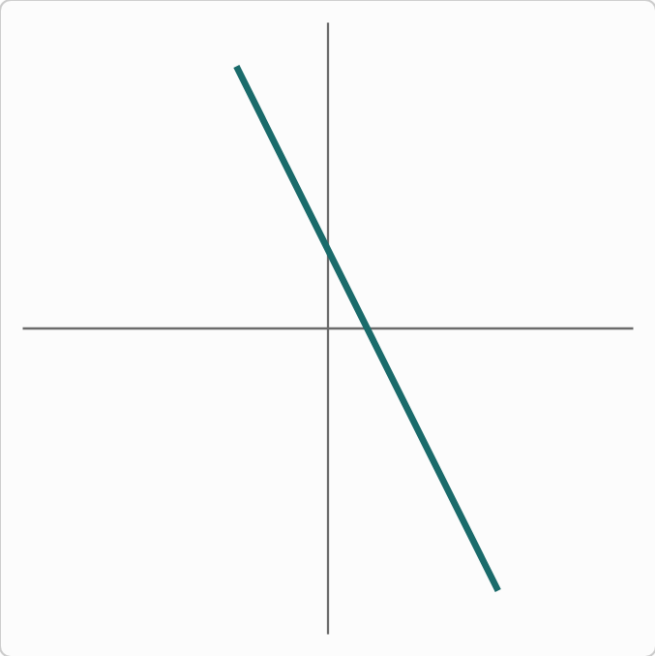
Question

Does this equation define a function of x?

Answer and reasoning

Here $2y^4=-3x^5$. Raising to an even power loses the sign: for $x=-1$, $y^4=3/2$, and both $y=\sqrt{(4]\{3/2)}$ and $y=-\sqrt{(4]\{3/2)}$ work. So one value of x gives two values of y. So **this equation does not define a function of x.**

Example 6: Graph 1



Plotted data: a straight line passing through points including *(0, 3)* and *(1, 1)*, extending from the upper-left to the lower-right of the grid with no breaks or turns.

Question

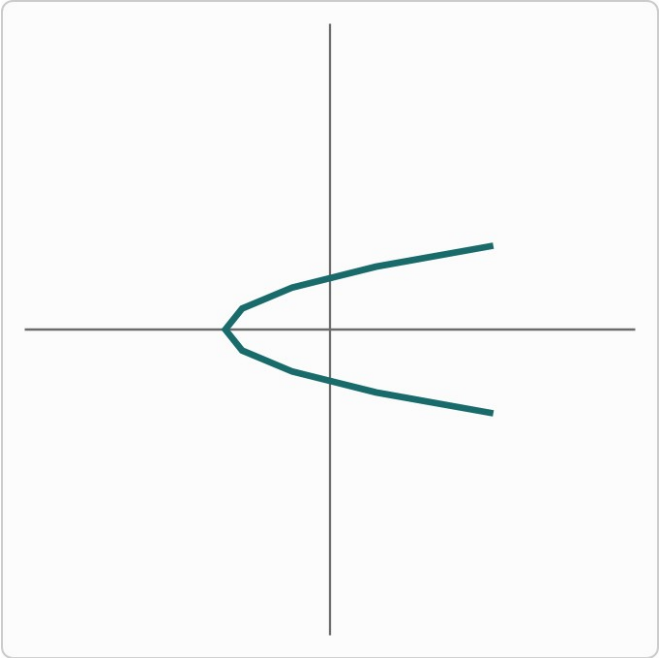
Does this graph represent a function?

Answer and reasoning

This is the line $y=-2x+3$. Any vertical line you draw crosses a straight (non-vertical) line exactly once. **This graph represents a function.**

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Example 7: Graph 2



Plotted data: a smooth curve shaped like a parabola opening to the right. Its leftmost point is *(−4, 0)*. One branch rises toward the upper right, reaching about *(1.8, 2.4)* at the edge of the grid, and the other falls toward the lower right, reaching about *(1.8, −2.4)*.

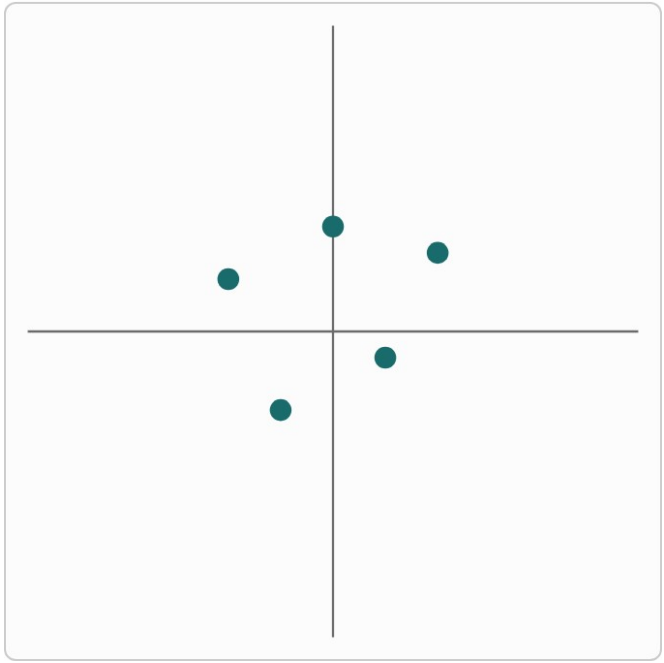
Question

Does this graph represent a function?

Answer and reasoning

This is the graph of $x=y^2-4$ — a parabola opened sideways instead of up. A vertical line through the open side of the curve crosses it twice (once above the x-axis, once below). **This graph does not represent a function.**

Example 8: Graph 3



Plotted data: five isolated points, not connected to each other, at approximately *(−4, 2)*, *(−2, −3)*, *(0, 4)*, *(2, −1)*, and *(4, 3)*.

Question

Does this graph represent a function?

Answer and reasoning

Check the x-coordinates: *−4, −2, 0, 2, 4* — all five are different, so no vertical line can pass through more than one point. **This graph represents a function.**

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Example 9: Slip 1

Table 2. Table of x-values and y-values, listed in two rows

x	2	4	6	8
y	5	5	5	5

Question

A student says this table is not a function, because the y-value 5 repeats. What is the mistake, if any?

A. Only an input with two different outputs breaks a function.
B. The table needs more rows before anyone can decide either way.
C. A function has to use each output only once, so this is wrong.
D. There is no mistake.

Answer and reasoning

The question is whether one input ever has two different outputs. Each x-value here, 2, 4, 6, and 8, appears once, so each input has exactly one output. Many inputs sharing the output 5 is allowed. **The table is a function; repeated y-values do not matter.**

Example 10: Slip 2

(1, 4), *(2, 4)*, *(3, 4)*, *(4, 9)*

Question

A student says these ordered pairs define a function, because no x-value repeats, even though 4 repeats as a y-value. What is the mistake, if any?

A. The y-value 4 repeats, so these are not a function.
B. A function needs an x-value of 0 among its pairs.
C. There is no mistake.
D. The pairs have to be in order of y to be checked.

Answer and reasoning

The x-values 1, 2, 3, and 4 are all different, so no input has two outputs. Several inputs sharing the output 4 is fine. The student checked the right thing. **There is no mistake: the pairs define a function.**

Example 11: Slip 3

$$x^2 + y = 4$$

Question

A student says this is not a function, because x is squared. What is the mistake, if any?

A. There is no mistake.
B. Squaring x is fine; a squared y is what gives two outputs.
C. It is not a function, because its graph is a curve, not a line.
D. It is a function only when x is a positive number.

Answer and reasoning

Solve for y: $y=4-x^2$. Each x gives exactly one y; for example $x=2$ and $x=-2$ both give $y=0$, which is two inputs sharing an output, not one input with two outputs. In equations like CheckIt's, where y appears in one term, the power that matters is the power on y. **The equation defines a function.**

Example 12: Slip 4

$$y^2 = x + 1$$

Question

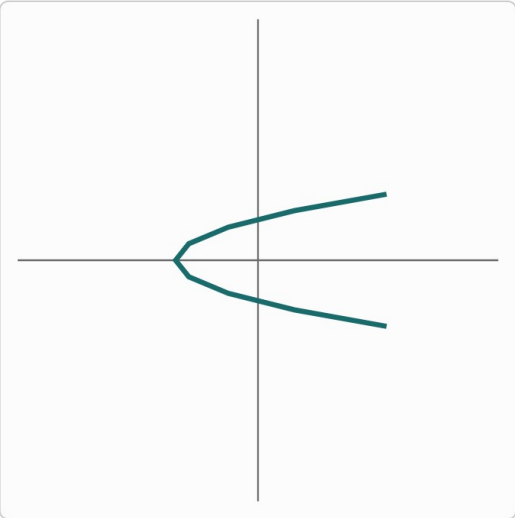
A student says this is a function, because x only has a power of 1. What is the mistake, if any?

A. It is a function, but only for x values greater than 0.
B. There is no mistake.
C. With y squared, one x can give two values of y.
D. The power on x has to be 2 before it can be a function.

Answer and reasoning

The power on y is even. Try $x=3$: $y^2=4$, so $y=2$ or $y=-2$. One input gives two outputs.
The equation does not define a function.

Example 13: Slip 5



Plotted data: a smooth curve shaped like a parabola opening to the right. Its leftmost point is *(−4, 0)*. One branch rises toward the upper right, reaching about *(1.8, 2.4)* at the edge of the grid, and the other falls toward the lower right, reaching about *(1.8, −2.4)*.

Question

A student says this graph is a function, because a horizontal line crosses it only once. What is the mistake, if any?

A. There is no mistake.
B. A horizontal line crosses it twice, so it is not a function.
C. A curve can never be a function, only a straight line can.
D. Use a vertical line: at x = 1.8 the curve is hit twice.

Answer and reasoning

Functions are checked with a vertical line, because a vertical line is one input. At $x=1.8$ the curve has two points, *(1.8, 2.4)* and *(1.8, −2.4)*: one input, two outputs. A horizontal line test answers a different question. **The graph is not a function.**